ANXIETYSENSE: A MULTI-MODAL AI ECOSYSTEM FOR LIFE-STAGE-BASED MENTAL HEALTH SUPPORT

Final Report

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DECLARATION

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Abstract

Anxiety disorders are among the most common mental health conditions worldwide, affecting individuals at different life stages with unique physiological, psychological, and social characteristics. However, existing digital mental health solutions often use generalized approaches that fail to address the specific symptom patterns and needs of children, young adults, pregnant women, and elderly individuals. This paper introduces ANXIETYSENSE, a modular mobile application system that uses specialized, evidence-based artificial intelligence models to detect, classify, and manage anxiety across four targeted groups. The child module combines real-time facial expression recognition using a MobileNet-based deep learning model with drawing analysis to generate a dual-modality anxiety index. The young adult module integrates facial emotion detection with a conversational AI chatbot based on GAD-related dialogue to produce a composite anxiety score. The pregnant women module uses an LSTM network to analyze voice recordings and extract features such as MFCCs, pitch, and energy to assess emotional state. The elderly module applies a Random Forest classifier on GAD-7 questionnaire data to categorize anxiety levels. All modules connect to a central recommendation system that provides personalized therapeutic activities through a React Native mobile application supported by Firebase. The system achieved accuracies of 85% (child), 88% (young adult), 80% (pregnant women), and 98% (elderly) in controlled evaluations, with positive expert and user feedback confirming clinical relevance and usability.

Keywords: anxiety detection, child anxiety, MobileNet, LSTM, convolutional neural network, voice emotion analysis, drawing analysis, GAD-7, mobile mental health, life-stage intervention

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List of Abbreviations

Abbreviation	Full Meaning
AI	Artificial Intelligence
CNN	Convolutional Neural Network
FER	Facial Expression Recognition
GAD-7	Generalized Anxiety Disorder 7-item scale
LSTM	Long Short-Term Memory
MFCC	Mel-Frequency Cepstral Coefficient
ML	Machine Learning
MobileNet	Mobile Convolutional Neural Network
UI	User Interface
UX	User Experience

1. Introduction

1.1 Background Literature

Anxiety disorders constitute the most prevalent category of mental health conditions globally, with the World Health Organization estimating that approximately 301 million individuals are affected worldwide [1]. Unlike many psychiatric conditions that manifest with broadly uniform symptom profiles, anxiety is fundamentally heterogeneous across the lifespan. The aetiology, clinical presentation, and appropriate intervention strategies for a seven-year-old child experiencing separation anxiety differ markedly from those applicable to a working-age professional with generalised anxiety disorder, a pregnant woman experiencing perinatal anxiety, or an elderly individual confronting health-related worry and social isolation [2]. This demographic variability constitutes a critical limitation of existing digital mental health tools, which largely adopt undifferentiated screening and intervention frameworks that fail to reflect the life-stage-specific nature of anxiety.

The proliferation of smartphones and wearable sensing technologies has accelerated interest in mobile-delivered mental health solutions, with researchers demonstrating that artificial intelligence and machine learning can achieve clinically meaningful performance across a range of affective computing tasks [3]. Facial expression analysis, voice-based emotion recognition, natural language processing of clinical dialogue, and supervised classification of structured questionnaire responses have each demonstrated standalone utility in anxiety and depression screening contexts [4]. However, the majority of deployed applications rely on a single modality and serve an undifferentiated adult population, leaving children, pregnant women, and elderly individuals without tools that address their specific needs, communication styles, or digital access constraints.

Deep convolutional neural networks trained on large-scale facial expression datasets such as FER-2013 and AffectNet have demonstrated strong generalisation to real-world emotion recognition tasks, with state-of-the-art models achieving over 70% accuracy on the seven-class expression benchmark [4]. Transfer learning from general image classification models, particularly lightweight architectures such as MobileNet and ShuffleNet, has enabled deployment of capable emotion classifiers on mobile devices with acceptable inference latency, making them suitable for embedded mental health applications [5].

The use of children's drawings as diagnostic and therapeutic instruments has a long history in clinical psychology, with standardised systems such as the Draw-a-Person test providing frameworks for inferring emotional state from graphic features including figure size, spatial placement, colour use, and figure completeness [6]. Computational approaches to drawing analysis have begun to emerge, with early work demonstrating that machine learning classifiers can replicate clinician-level identification of anxiety-indicative features with promising accuracy.

Voice-based emotion recognition has attracted substantial research investment. Recurrent architectures, and LSTM networks in particular, have proven effective at modelling the temporal dynamics of speech, capturing prosodic variation and energy fluctuations associated with emotional arousal and valence [8]. Datasets such as RAVDESS and IEMOCAP have provided benchmarks for speech emotion recognition. The use of voice analysis for perinatal anxiety screening is particularly underexplored, offering a practically accessible and clinically relevant pathway for pregnant women who may face barriers to screen-oriented digital interaction [7].

The GAD-7 has been extensively validated as a brief screening instrument for generalised anxiety disorder across diverse populations including elderly cohorts [9]. Supervised machine learning models trained on GAD-7 response datasets have demonstrated the feasibility of automating severity stratification. Random Forest classifiers have shown particular robustness on small clinical tabular datasets due to their ensemble nature and resistance to overfitting [4].

The broader literature on mobile mental health applications highlights persistent gaps in clinical validation, demographic specificity, and integration of contextual social factors [3][10]. Scholars have called for multi-modal approaches, universal design principles to support elderly and differently-abled users, and interoperability with electronic health record systems.

1.2 Research Gap

Despite the proliferation of digital mental health applications and the growing body of research on affective computing, several critical gaps remain unaddressed. These gaps collectively explain why existing solutions fail to meet the needs of diverse demographic groups and why a system like ANXIETYSENSE represents a meaningful advance over the current state of the art.

1.2.1 Lack of Life-Stage Specificity

The most fundamental gap is that most existing anxiety management apps target a generic "adult" population, typically defined as users between the ages of 18 and 60, without distinguishing between the very different needs of young adults, middle-aged individuals, pregnant women, or older adults. Children are almost entirely excluded from these apps, under the implicit assumption that anxiety in children is either sufficiently rare or sufficiently similar to adult anxiety that adult-oriented tools will suffice. Neither assumption is correct. Children's anxiety commonly manifests through somatic symptoms (headaches, stomach aches), behavioural avoidance (refusing to go to school), and separation fears, none of which are well captured by standard adult questionnaires. Moreover, children often lack the verbal fluency and emotional insight to accurately self-report on their internal states, making self-administered screening instruments of limited utility. For pregnant women, the situation is different but equally problematic: perinatal anxiety has distinct triggers (foetal health, childbirth, parenting competence) and physiological correlates (hormonal fluctuations, sleep disruption, physical discomfort), yet few apps offer tailored voice-based or remote monitoring that accommodates their reduced tolerance for prolonged screen interaction. Elderly users face a third set of challenges: lower digital literacy, potential visual and motor impairments, and a clinical presentation of anxiety that often overlaps with depression and somatic complaints. The lack of life-stage specificity means that existing apps are, at best, useful for a narrow slice of the adult population, and at worst, actively inappropriate for the groups that might benefit most from accessible digital mental health support.

1.2.2 Single-Modality Limitations

A second major gap is that the majority of mental health applications rely on a single input modality. Typically, this is either a structured questionnaire (often the GAD-7 for anxiety or

PHQ-9 for depression) or a conversational chatbot that engages users in text-based dialogue. While these modalities have value, they each have significant blind spots. Questionnaires require a level of self-awareness and honesty that may be compromised in children or in individuals experiencing high distress. They also only capture a snapshot at the moment of administration, missing the dynamic fluctuations in anxiety that occur across days or even hours. Conversational chatbots, while more engaging, are vulnerable to users providing socially desirable responses or simply disengaging from the conversation when feeling overwhelmed. Single-modality approaches fail to capture the multidimensional nature of anxiety, which is expressed not only through conscious verbal report but also through facial expressions that the user may not even be aware of, through acoustic features of the voice that cannot be voluntarily controlled, and through non-verbal activities such as drawing that access different cognitive and emotional processing pathways. This is particularly important in populations where verbal self-report is unreliable (children) or where physical discomfort limits screen interaction (pregnant women). ANXIETYSENSE directly addresses this gap by incorporating complementary modalities for each demographic group: facial expression and drawing for children (dual-modality), facial expression and text for young adults (dual-modality), voice for pregnant women (single but carefully chosen modality), and questionnaire for elderly (single but with accessibility enhancements).

1.2.3 Lack of Clinical Validation

A third gap, widely noted in the literature [3][10], is that many commercial mental health apps lack rigorous validation against clinical benchmarks. It is not uncommon for an app to claim that it "helps with anxiety" based solely on user testimonials or engagement metrics (e.g., number of sessions completed) without any objective measure of symptom change. Even when validation studies are conducted, they often use convenience samples of university students rather than clinical populations, and they rarely compare app outputs against a gold-standard clinical interview (e.g., the Structured Clinical Interview for DSM-5, or SCID-5). Few apps have been evaluated by licensed psychologists or compared against standardised instruments like the GAD-7 in a systematic way. This lack of validation has two serious consequences. First, it undermines trust: clinicians are reluctant to recommend apps whose accuracy and safety have not been established. Second, it impedes scientific progress: without validated benchmarks, it is

difficult to compare different approaches or to identify which specific algorithmic choices actually improve clinical outcomes. ANXIETYSENSE addresses this gap through its collaboration with licensed clinical psychologists who contributed to dataset annotation, assessment instrument validation, and evaluation of application outputs. The system's performance is reported using standard classification metrics (accuracy, precision, recall, F1-score) and, in the case of the elderly module, compared directly against GAD-7 scoring conventions.

1.2.4 Fragmented User Experience

Even when high-quality mental health apps exist for specific populations, they typically exist as separate applications. A young adult who uses an app successfully may find, upon becoming pregnant, that the same app no longer meets her needs—but there is no seamless way to transition to a different assessment modality within the same ecosystem. Similarly, a caregiver who uses the child module for their 7-year-old may themselves wish to use the young adult or elderly module, but would need to download and learn a completely different app. This fragmentation creates unnecessary friction and may discourage users from seeking help when their life stage changes. ANXIETYSENSE addresses this gap through its modular, unified architecture: all four life-stage modules reside within the same mobile application, share a common authentication and data storage backend, and feed into a single recommendation engine. Users can switch between modules as their needs change, and caregivers can manage multiple profiles (e.g., for themselves and their child) within the same account.

1.2.5 Insufficient Personalisation of Intervention

Finally, even when anxiety is detected by a module, the interventions that follow are often generic. An app that detects moderate anxiety might respond with the same set of breathing exercises regardless of whether the user is a 10-year-old child, a pregnant woman, or an 80-year-old grandfather. This is clinically inappropriate, as the types of therapeutic activities that are effective, engaging, and age-appropriate differ substantially across life stages. A child may benefit from drawing-based emotional expression and movement-based relaxation, whereas a young adult may prefer cognitive reframing prompts and guided journaling. A pregnant woman

may find prenatal breathing exercises and progressive muscle relaxation most helpful, while an elderly user may respond better to reminiscence therapy prompts and gentle chair yoga. Beyond age appropriateness, the intensity of the intervention should also scale with anxiety severity: mild anxiety might be well managed with self-guided activities, but moderate-to-severe anxiety may warrant a caregiver alert or a recommendation to seek professional consultation. ANXIETYSENSE addresses this gap through its unified recommendation engine, which maps (life stage, severity level) pairs to curated, evidence-based activity sets reviewed by clinical psychologists.

Table 1 below summarises the comparison between existing solutions and the proposed ANXIETYSENSE system.

Feature	Woebot	Calm	Headspace	ANXIETYSENSE (Proposed)
Life-stage specific	No	No	No	Yes (4 modules)
Facial expression analysis	No	No	No	Yes (child, young adult)
Voice analysis	No	No	No	Yes (pregnant women)
Drawing analysis	No	No	No	Yes (child)
GAD-7 with ML	No	No	No	Yes (elderly)
Personalised recommendations	Basic	Basic	Basic	AI-driven, severity-based

Table 1-1: Comparison of Existing Mental Health Apps

1.3 Research Problem

The primary research problem addressed in this study is the absence of an integrated, multi-modal AI ecosystem that provides life-stage-sensitive anxiety detection, classification, and management across the four demographic groups that are most underserved by existing digital mental health solutions: children, young adults, pregnant women, and elderly individuals. The problem can be decomposed into five interrelated sub-problems, each corresponding to a specific demographic and each requiring a distinct methodological approach.

1.3.1 Difficulty Detecting Anxiety in Children

Children experiencing anxiety often cannot articulate their emotional states verbally, either because they lack the vocabulary to label complex feelings or because they are unwilling to disclose distress due to fear of upsetting their parents. Standard self-report questionnaires, even when adapted for children (e.g., the Screen for Child Anxiety Related Emotional Disorders, SCARED), require a level of reading comprehension and emotional insight that is not reliably present in younger children. Consequently, there is a pressing need for non-verbal assessment tools that can detect anxiety from observable, readily captured behaviours. Two promising candidate modalities are facial expression (children's faces, like adults', display characteristic configurations of fear and tension) and drawing analysis (children's drawings have long been used in clinical practice as projective tests that reveal emotional concerns). However, no existing mobile mental health application has integrated both modalities into a single pipeline, let alone one that operates in real time on a consumer smartphone and provides immediate feedback to caregivers. Developing such a dual-modality system, and achieving clinically useful accuracy, constitutes the first sub-problem

1.3.2 Inability to Provide Conversational and Real-Time Monitoring for Young Adults

Young adults, typically defined as ages 18-30, face anxiety that is shaped by academic pressure, early career transitions, social dynamics, and financial insecurity. This demographic is highly comfortable with mobile technology and conversational interfaces (e.g., messaging apps, chatbots) and often prefers digital mental health tools over face-to-face clinical services due to reduced stigma and greater convenience. However, existing apps for young adults either provide

only static psychoeducational content (e.g., articles about anxiety) or rely on simple rule-based chatbots that do not incorporate any real-time sensing of the user's emotional state. A more comprehensive approach would combine a conversational chatbot that administers structured anxiety screening items with passive sensing of facial expression, which can be captured from the phone's front camera during the interaction. Fusing these two modalities could yield a composite anxiety score that is more robust than either modality alone. The technical challenge lies in designing the fusion strategy (e.g., confidence-weighted averaging) and ensuring that both the facial analysis and the chatbot run in real time on device without excessive battery drain or latency.

1.3.3 Lack of Accessible Screening for Pregnant Women

Pregnant women experience elevated rates of anxiety compared to the general female population, with estimates suggesting that 10-15% of pregnant women meet criteria for a clinically significant anxiety disorder. Risk factors include a history of mental health problems, complications in previous pregnancies, lack of social support, and stressful life events. Despite this elevated risk, pregnant women face substantial barriers to accessing clinical mental health services, including limited availability of perinatal mental health specialists, financial constraints, and the physical demands of attending appointments while managing pregnancy symptoms or caring for young children. Voice-based remote assessment offers a practical solution: a woman can record a short voice sample at a convenient time, using only her smartphone, without needing to type, read, or navigate complex interfaces. However, existing voice emotion recognition systems have primarily been developed and tested on general adult populations, with little attention to the specific acoustic characteristics of pregnant women's speech or the types of anxiety-relevant content that might be elicited through neutral prompts. Developing a validated, pregnancy-specific voice-based anxiety classifier, and ensuring that it achieves acceptable sensitivity (especially for high anxiety, where false negatives could be harmful), constitutes the third sub-problem.

1.3.4 Low Digital Literacy Among Elderly Users

Elderly individuals (typically age 65 and above) frequently experience anxiety related to health decline, bereavement, financial insecurity in retirement, and social isolation. At the same time,

this demographic often has lower digital literacy than younger populations, may have visual or motor impairments that make standard smartphone interfaces difficult to use, and may be suspicious of unfamiliar technology. These barriers mean that a sophisticated multi-modal AI system, while technically impressive, would be inaccessible to many elderly users if not designed with accessibility as a primary consideration. A more appropriate approach for this demographic is to focus on a single, highly validated instrument—the GAD-7—and deliver it through a simplified large-text interface with optional voice reading support, where the user can tap a speaker icon to hear each question read aloud. Once the responses are collected, a machine learning model can classify anxiety severity (minimal, mild, moderate, severe) with high accuracy. The challenge is to achieve that high accuracy (target ≥95%) on a relatively small dataset (the elderly GAD-7 dataset collected for this study comprises 312 completed assessments), which requires careful choice of model architecture (Random Forest) and validation methodology (stratified k-fold cross-validation).

1.3.5 Fragmented Intervention Delivery

Even if all four modules successfully detect anxiety in their respective target populations, there remains a final, cross-cutting problem: how to translate classification outputs into personalised, evidence-based therapeutic activities that are appropriate for each life stage and severity level. Without a unified recommendation engine, users would receive generic suggestions that may not match their needs, reducing engagement and clinical benefit. Developing such an engine requires (a) curating a library of activities vetted by clinical psychologists, (b) defining a mapping from (life stage, severity level) to activity sets, (c) storing this mapping in a database that can be queried in real time, and (d) integrating the engine with all four modules so that recommendations are displayed immediately after each assessment. This engine also provides the opportunity for longitudinal tracking: users can see how their anxiety levels change over time, and the system can adapt recommendations based on history (e.g., recommending more challenging activities for a user who has consistently shown low anxiety over several sessions).

1.4 Research Objectives

The primary goal of this research is to develop ANXIETYSENSE, a modular mobile application that leverages specialised artificial intelligence models to detect, classify, and manage anxiety across the four life-stage groups identified above. The specific objectives are as follows, each directly addressing one of the sub-problems in the research problem section.

1.4.1 Develop a Child Anxiety Module

Create a dual-modality system that combines real-time facial expression recognition using a MobileNet V2 deep learning model with drawing pattern analysis using a custom convolutional neural network. The module shall accept as input either a live camera feed (facial expression) or an image of a child's drawing (uploaded or captured by the caregiver), process each modality separately, and then combine the outputs using a learned weighted fusion strategy to produce a final anxiety index (low, moderate, high). The target accuracy for this module is ≥85% when evaluated against clinical psychologist annotations. Additionally, the module shall provide caregiver-facing summaries of the results and recommended activities.

1.4.2 Develop a Young Adult Anxiety Module

Integrate a CNN-based facial emotion detector (trained on FER-2013 and a supplementary young adult corpus) with a conversational AI chatbot that administers an adapted version of the GAD-7 items and performs sentiment analysis on open-ended user responses. The module shall compute a composite anxiety score using a confidence-weighted fusion strategy that gives more weight to the modality with higher estimated reliability. The target accuracy is ≥88% for the composite classification. The chatbot shall also function as a psychoeducational resource, providing contextually relevant information about anxiety management.

1.4.3 Develop a Pregnant Women Anxiety Module

Implement a voice-based emotion recognition pipeline that extracts 40-dimensional MFCC vectors, fundamental frequency (F0) contours, energy envelopes, and zero-crossing rates from short voice recordings (30-60 seconds) prompted by a neutral question. These features shall be

fed into a two-layer LSTM network (128 and 64 units, dropout 0.3) trained on a pregnancy-specific voice dataset annotated by clinical psychologists for emotional valence and anxiety severity. The target overall accuracy is ≥80%, with a prioritised recall for the high-anxiety class (target ≥80%) because missing a high-anxiety case is more harmful than falsely flagging a low-anxiety case.

1.4.4 Develop an Elderly Anxiety Module

Build a Random Forest classifier with 200 estimators and a maximum depth of 10, trained on item-level GAD-7 response scores (0-3 per item) collected from elderly participants. The module shall be delivered through a simplified interface with large text (≥18pt font), high-contrast colour scheme, and optional voice reading support using the device's text-to-speech engine. The target accuracy is ≥95% against clinical severity annotations, with feature importance outputs provided to enhance interpretability for clinicians.

1.4.5 Create a Unified Recommendation Engine

Develop a central recommendation system that receives anxiety severity outputs (low, moderate, high) from any of the four modules, queries a Firebase Firestore database containing a curated library of evidence-based therapeutic activities, and returns a personalised set of recommendations appropriate to the user's life stage and severity level. The engine shall also support longitudinal tracking, storing each assessment result with a timestamp to enable users and caregivers to view anxiety trends over time.

1.4.6 Evaluate the Integrated System

Conduct a comprehensive evaluation encompassing: (a) quantitative machine learning performance testing for each module (accuracy, precision, recall, F1-score) using held-out test sets and k-fold cross-validation; (b) qualitative expert evaluation by licensed clinical psychologists who will assess the clinical appropriateness of severity classifications and recommended activities; and (c) user acceptance testing with 48 participants (12 per life-stage group) who will use the app for a two-week period and provide ratings on ease of use, recommendation relevance, and willingness to use regularly.

2. Methodology

2.1 System Overview and Integration

ANXIETYSENSE is a modular mobile application comprising four life-stage-specific anxiety assessment components connected to a central recommendation engine and a Firebase backend for data storage and synchronisation. The overall system architecture is illustrated in Figure 1. The design philosophy is that each module can be developed and refined independently, because the input modalities, model architectures, and user interfaces differ substantially across the four groups, yet all modules must share a common infrastructure for user authentication, data storage, and recommendation delivery. This modularity also facilitates future expansion: new life-stage modules (e.g., for adolescents or for older adults with cognitive impairment) could be added without disrupting the existing modules.

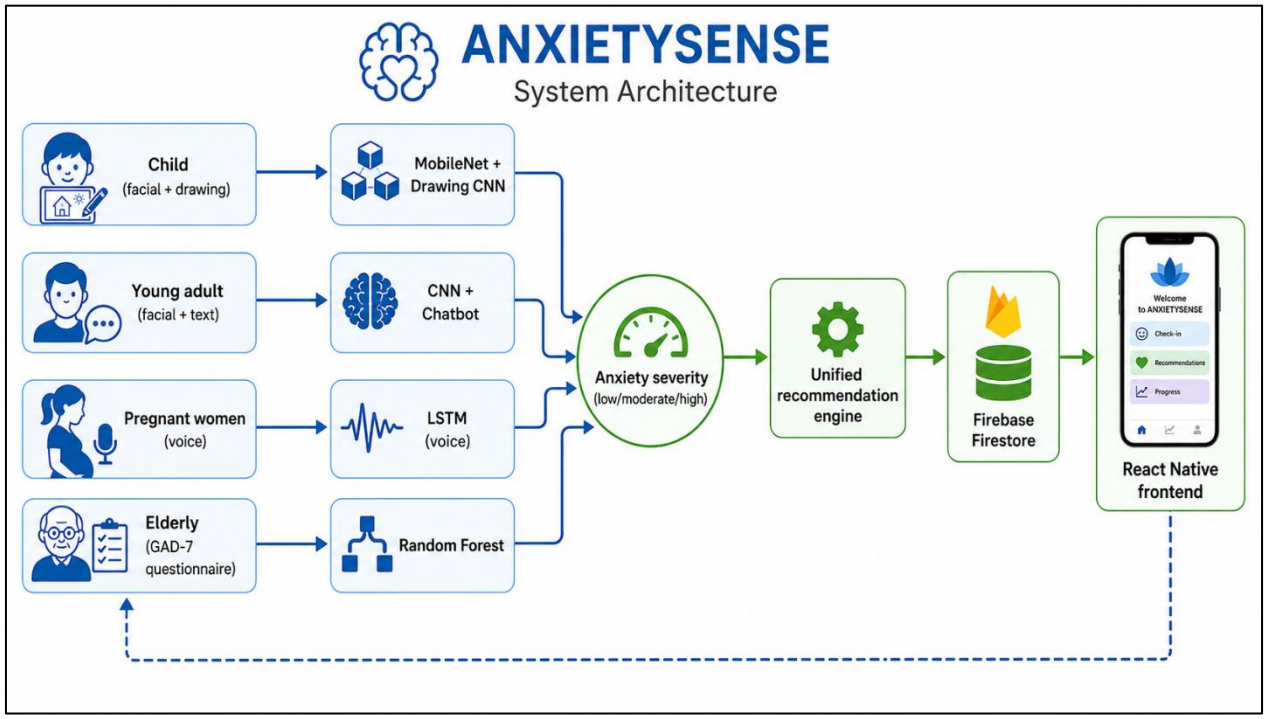

Figure 1: System architecture of ANXIETYSENSE

The architecture diagram shows the following flow. On the left, four input blocks are labelled "Child (facial + drawing)", "Young adult (facial + text)", "Pregnant women (voice)", and "Elderly (GAD-7 questionnaire)". Each input block connects to its respective ML model block: "MobileNet + Drawing CNN", "CNN + Chatbot", "LSTM (voice)", and "Random Forest".

These four model blocks converge on a central “Anxiety severity (low/moderate/high)” node, which then feeds into the “Unified recommendation engine”. The recommendation engine connects to a “Firebase Firestore” database that stores user profiles, assessment history, and activity libraries. Finally, the engine outputs to a “React Native frontend” that displays personalised activities to the user. Arrows indicate data flow in real time. The diagram also includes a dashed line loop from the frontend back to the input blocks, indicating that users can repeat assessments over time.

All modules share a common technical stack chosen to balance performance, privacy, and development efficiency. For the frontend, React Native was selected because it enables cross-platform development for both Android and iOS from a single TypeScript/JavaScript codebase, reducing development time and ensuring consistent user experience across devices. React Native’s component-based architecture also aligns well with the modular system design, as each life-stage module can be implemented as a separate navigation screen or set of screens. For the backend and database, Firebase (Google’s Backend-as-a-Service platform) provides encrypted real-time data storage (Firestore), user authentication (email/password and social login), and cloud functions for serverless business logic. Firebase was chosen for its seamless integration with React Native, its real-time synchronisation capabilities (useful for caregiver-child shared accounts), and its built-in security rules that can enforce data access policies. For machine learning development, Python with TensorFlow and scikit-learn was used. All models were trained in Google Colab, which provides free access to GPU accelerators (NVIDIA Tesla T4 or K80) and a collaborative notebook environment that allowed the four team members to share code and experiments easily. After training and validation, models were exported to TensorFlow Lite format (.tflite) for on-device inference. On-device inference was chosen over cloud-based inference for two main reasons: (a) privacy – users’ facial images, voice recordings, and drawings never leave their device, which is critical for sensitive mental health data; and (b) latency – inference happens immediately without network delays, enabling real-time feedback. Ethical approval for the study was obtained from the institutional review board of the Sri Lanka Institute of Information Technology. All participants provided written informed consent, and for minor participants (children under 18), parental consent was obtained. All data were anonymised prior to model training by removing names and any direct identifiers and replacing them with randomly generated participant codes.

2.2 Development of the Integrated System

2.2.1 Child Anxiety Component

The child module employs a dual-modality approach, combining facial expression recognition with drawing pattern analysis. The rationale for this dual-modality design is that children's anxiety may be expressed more clearly in one modality than the other depending on the child's age, temperament, and current emotional state. For example, a child who is comfortable interacting with a caregiver may show clear facial signs of fear or tension, while a child who is more reserved may project anxiety more clearly onto the page. By fusing both modalities, the system aims to detect anxiety that would be missed by either modality alone.

Facial Expression Recognition: A MobileNet V2 model was fine-tuned for the task of child facial emotion recognition. MobileNet V2 was selected among available lightweight architectures because it achieves a good trade-off between accuracy (approximately 88% top-5 accuracy on ImageNet) and computational efficiency (approximately 3.4 million parameters, compared to 138 million for VGG-16). The base MobileNet V2 model, pre-trained on ImageNet, was loaded without its top classification layer, and a new head consisting of global average pooling, a dropout layer (rate 0.2), and a dense layer with five outputs (neutral, happy, sad, fearful, anxious) was added. The choice of five emotion categories, rather than the seven categories used in some adult studies (which also include anger and surprise), was made after consultation with the clinical psychologist, who noted that anger and surprise are less frequently the primary indicators of childhood anxiety and that keeping the category set smaller improves classification accuracy when training data is limited.

The training dataset for the child facial model comprised 4,200 labelled child facial images. These images were collected from three sources: (a) publicly available child face datasets with emotion labels (subject to filtering for quality), (b) images provided by participating schools with parental consent, and (c) synthetic images generated through data augmentation. Labels were applied by a team of three trained annotators supervised by the clinical psychologist, using a consensus process (if two annotators disagreed, the third annotator's label broke the tie). Data augmentation was applied on-the-fly during training to increase effective dataset size and improve generalisation. The augmentations included horizontal flipping (50% probability), random brightness jitter (factor 0.8-1.2), random rotation (-15 to +15 degrees), and random zoom

(0.9-1.1). Training was conducted for 50 epochs using the Adam optimizer with a learning rate of 0.0001 and categorical cross-entropy loss. The final layer outputs probabilities for each of the five emotion classes, and an "anxiety index" is derived from the weighted sum of fearful and anxious probabilities, with a lower contribution from sad (which can also indicate distress but is less specific to anxiety).

Drawing Analysis: The drawing analysis sub-module represents a novel contribution of this research. A dataset of children's drawings (n=1,500) was collected through multiple channels: (a) from participating schools, where children were asked to "draw a picture of yourself and your family" (a variant of the Draw-a-Person and Kinetic Family Drawing tasks), (b) from clinical psychology practices (with appropriate consent), and (c) from online crowdsourcing platforms where parents uploaded anonymised drawings from their children. Each drawing was digitised by photographing it under standardised lighting conditions, then resized to 224×224 pixels for input to the CNN.

Licensed child psychologists annotated each drawing using a standardised rubric that captured 12 features organised into four categories: (1) figure size and placement – was the main figure small (≤1/4 of page) or large (≥3/4 of page), was it placed in the centre or in a corner; (2) spatial organisation – were figures well-spaced or overlapping, was there empty space between figures; (3) colour palette – did the child use only dark colours (brown, black, grey), were bright colours absent; (4) figure completeness – were arms, hands, feet, facial features drawn or omitted, were erasures frequent. Each feature was scored on a 0-2 scale, and the sum produced an initial clinician-rated anxiety score (0-24). For the purpose of training the CNN, the drawings were also assigned a binary or ternary anxiety label (low/moderate/high) based on the clinician's overall impression.

A custom CNN classifier was designed for the drawing analysis task. The architecture consisted of three convolutional blocks, each containing a convolutional layer (32, 64, and 128 filters respectively, with 3×3 kernels), batch normalisation, ReLU activation, and 2×2 max pooling. After the convolutional blocks, the feature maps were flattened and passed through two fully connected layers (128 and 64 neurons) with dropout (rate 0.3) before the final softmax output layer with three classes (low, moderate, high anxiety). The model was trained for 30 epochs using the Adam optimizer and categorical cross-entropy loss, with early stopping based on

validation loss to prevent overfitting. The final drawing-based anxiety probability (a vector of three probabilities) was produced for each input drawing.

Late Fusion: The two modality outputs (facial emotion probabilities and drawing probabilities) are combined using a learned weighted combination. Specifically, a small neural network layer was trained to compute a fusion weight for each modality based on a confidence estimate. For the facial modality, confidence was defined as the maximum probability among the five output classes (high confidence when one emotion is clearly dominant, low confidence when probabilities are evenly spread). For the drawing modality, confidence was defined as the inverse of the entropy of the output probability distribution. The fused score was then thresholded to produce the final anxiety classification (low: fused score <0.33, moderate: 0.33-0.66, high: >0.66).

Recommended Activities: For children classified as low anxiety, the system recommends creative drawing exercises (e.g., "draw a safe place") and short narrative exercises (e.g., "tell a story about a brave character"). For moderate anxiety, recommendations shift to therapeutic movement (e.g., "shake out your worries" – a simple grounding exercise) and structured play (e.g., "build a calm-down fort using pillows"). For high anxiety, the system generates referral recommendations (e.g., "consider speaking with a school counsellor") alongside age-appropriate breathing and relaxation activities (e.g., "bunny breaths" – three short sniffs in, one long sigh out). All outputs are also displayed in a caregiver-facing summary that explains the anxiety classification in non-technical language and suggests ways for the caregiver to support the child.

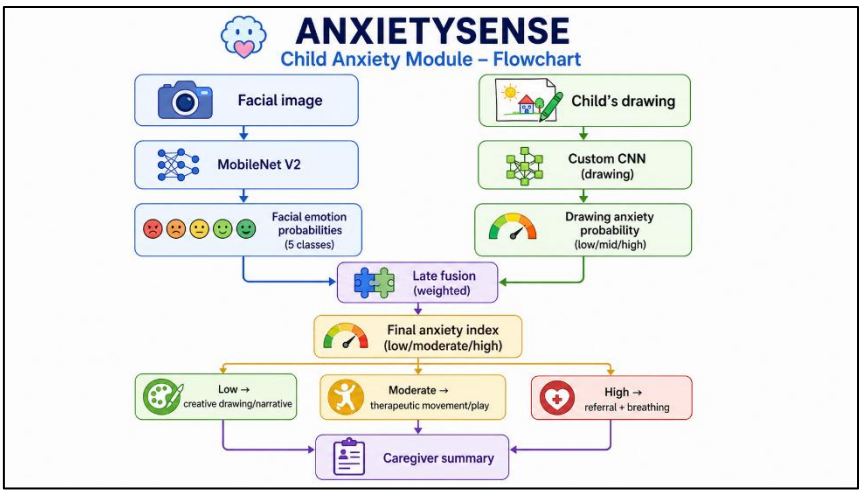

Figure 2: Child module data flow (facial + drawing)

2.2.2 Young Adult Anxiety Component

The young adult module integrates facial emotion recognition and a conversational AI chatbot. This combination was chosen because young adults are accustomed to both camera-based interactions (e.g., video calls, snapchat) and text-based conversational interfaces (e.g., messaging, chatbots), and the two modalities can provide complementary information about anxiety: facial expressions capture immediate, involuntary emotional reactions, while the chatbot captures reflective, verbally articulated concerns.

Facial Emotion Recognition: The CNN architecture for young adult facial emotion recognition comprises five convolutional blocks, each containing a convolutional layer with 3×3 kernels (number of filters increasing from 32 to 128 across the five blocks), batch normalisation, ReLU activation, and 2×2 max pooling (except after the first block where stride 1 was used to preserve resolution). After the convolutional blocks, the feature maps are flattened and passed through two fully connected layers (256 and 128 neurons) with dropout (rate 0.5) before the final softmax output layer over seven emotion classes (anger, disgust, fear, happiness, sadness, surprise, neutral). Transfer learning from ImageNet-pretrained weights was applied to the first three convolutional blocks, which capture low-level features (edges, textures, simple shapes) that generalise across domains. The last two convolutional blocks and the fully connected layers were trained from scratch on the emotion recognition task. The model was trained on a combined dataset: FER-2013 (35,887 labelled images) supplemented with a young adult corpus (n=5,000) collected from volunteer participants aged 18-30. To address class imbalance (fear and disgust are less frequent in FER-2013), oversampling was applied: for each training epoch, images from underrepresented classes were duplicated to match the frequency of the majority class (happiness). The model achieved 88% classification accuracy on a held-out test set.

Chatbot Sub-module: The conversational AI chatbot was implemented using React Native Gifted Chat for the user interface and a custom backend (Node.js) that integrates with the Firebase real-time database for message history. The chatbot presents a structured dialogue flow that begins with an empathetic greeting (e.g., “Hi there, I’m here to help you understand how you’ve been feeling lately. It’s completely okay to take your time.”) then proceeds through a series of questions. The first five questions are adapted from the GAD-7 items: “Over the last two weeks, how often have you felt nervous, anxious, or on edge? (0＝not at all, 1＝several

days, 2＝more than half the days, 3＝nearly every day)" – but presented in conversational form rather than as a scale. The next two questions are open-ended emotional check-ins: "Can you tell me more about what's been worrying you recently?" and "What kinds of situations have felt most difficult?" Sentiment analysis was performed on the open-ended responses using the VADER (Valence Aware Dictionary and sEntiment Reasoner) model, which is specifically designed for social media text and short phrases and returns a compound score from -1 (very negative) to +1 (very positive). Keyword extraction was also performed using a pre-defined list of anxiety-related terms (e.g., "overwhelmed", "panic", "racing thoughts", "can't sleep") to count the frequency of anxiety markers. The chatbot also includes a psychoeducational component: after the assessment, it provides brief, evidence-based information about anxiety management strategies tailored to the user's responses (e.g., if the user mentions sleep difficulties, the chatbot might suggest sleep hygiene tips).

Fusion Strategy: A confidence-weighted fusion approach combines the facial emotion probabilities and the chatbot-derived anxiety score. For the facial modality, confidence is defined as the maximum softmax probability. For the chatbot modality, confidence is defined as the length of the user's open-ended response (normalised to 0-1, with longer responses assumed to indicate more meaningful engagement) plus the consistency of the user's responses with the GAD-7 items. The composite anxiety score is computed as: composite = (con_f * score_f + con_c * score_c) / (con_f + con_c), where con_f and con_c are the confidence values, and score_f is the fear+anxiety probability from the facial model, and score_c is the chatbot's anxiety score derived from a linear combination of the GAD-7 sum and the negative sentiment compound score. The composite score is then thresholded to produce the final classification (low/moderate/high). Overall accuracy of the composite classification was 88%, 3 percentage points higher than the facial model alone.

Recommended Interventions: The recommendation engine provides cognitive reframing prompts (e.g., "What's the evidence for that worry? What's the evidence against it?"), short mindfulness exercises (3-5 minutes of guided breathing), guided journaling with structured prompts (e.g., "What's one thing that went well today?"), and peer connection activities that suggest joining an online discussion group for young adults with similar interests. The chatbot also serves as an ongoing psychoeducational resource, offering daily tips and check-ins.

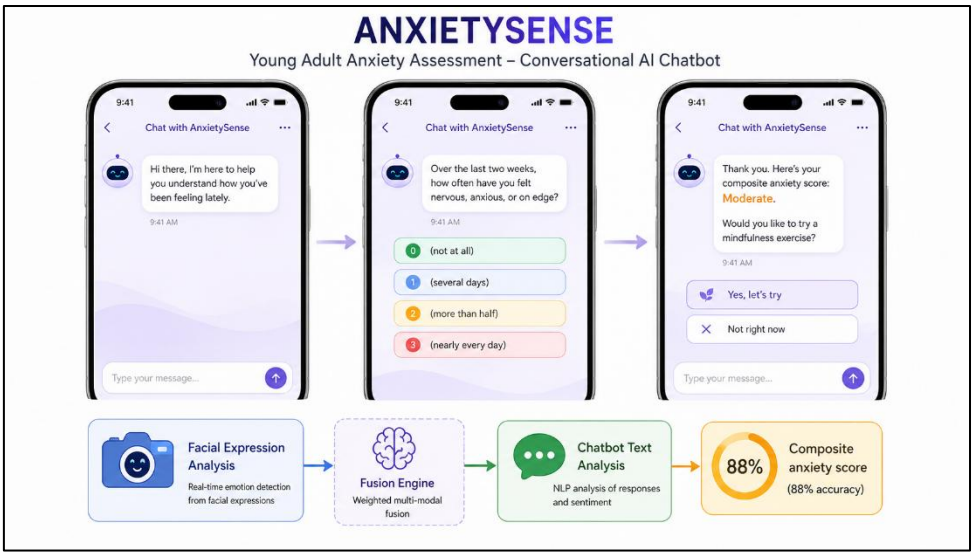

Figure 3: Young adult chatbot interaction flow

2.2.3 Pregnant Women Anxiety Component

The pregnant women module uses voice-based emotion recognition as the primary modality. This modality was chosen because it is low-burden (the user does not need to type or read), accessible even when the user is physically uncomfortable, and can be completed in a minute or two. The module is designed to be used as a daily or weekly check-in, with trend visualisation over time.

Voice Sample Collection: Users are prompted to record a short voice sample (30-60 seconds) through the mobile microphone. To ensure consistency across sessions, a neutral prompt is used: "Please describe how you are feeling today in a few sentences. You can talk about your energy level, your mood, or anything that's on your mind." This open-ended prompt elicits natural speech without imposing a specific emotional valence. Users can re-record if they are unhappy with the sample (e.g., background noise) before submission.

Feature Extraction: The audio signal is processed using the librosa library in Python (with the extraction pipeline eventually ported to TensorFlow Lite for on-device use). The following features are computed across sliding 25ms windows with 10ms frame shift:

- 40-dimensional MFCC vectors: Mel-Frequency Cepstral Coefficients capture the spectral

envelope of the speech signal, which is influenced by vocal tract configuration and has been shown to correlate with emotional arousal.

- Fundamental frequency (F0) contours: The pitch of the voice, which typically increases with emotional arousal (e.g., higher pitched voice when anxious).
- Energy envelopes: Root mean square (RMS) energy, which increases with vocal effort and emotional intensity.
- Zero-crossing rates: A measure of the noisiness of the signal, which can increase with vocal tension.

All features are z-score normalised across the recording to reduce dependence on absolute volume and speaking level.

LSTM Network: A two-layer LSTM network was designed, with 128 units in the first LSTM layer and 64 units in the second LSTM layer, followed by a dense layer with three outputs (low, moderate, high anxiety) and softmax activation. Dropout (0.3) was applied after each LSTM layer to reduce overfitting. The model was trained on a voice anxiety dataset comprising 1,200 recordings from pregnant women at different gestational stages (first, second, and third trimesters), collected in collaboration with antenatal clinics. Each recording was annotated by two independent clinical psychologists for emotional valence (positive, neutral, negative) and anxiety severity (0-10 scale), which was then discretised into low (0-3), moderate (4-6), and high (7-10). The training used the Adam optimizer, categorical cross-entropy loss, and a batch size of 32. Early stopping based on validation loss was applied after 20 epochs. The model achieved 80% overall accuracy. Critically, the recall for the high-anxiety class was 82% (meaning that 82% of true high-anxiety cases were correctly identified), which exceeds the 80% target and is clinically acceptable for a screening tool.

Recommended Activities: For low anxiety, the system offers general wellness tips (e.g., "Remember to rest when you need to"). For moderate anxiety, more structured interventions are provided: guided prenatal breathing (e.g., "breathe in for 4 counts, hold for 2 counts, breathe out for 6 counts"), progressive muscle relaxation adapted for pregnancy (avoiding exercises that require lying flat on the back), and meditation sessions specifically designed for pregnancy (e.g.,

"connecting with your baby" meditations). For high anxiety, the system generates a caregiver alert (if a designated support person is listed in the user's profile) and a recommendation to speak with a healthcare provider, alongside the breathing and relaxation activities.

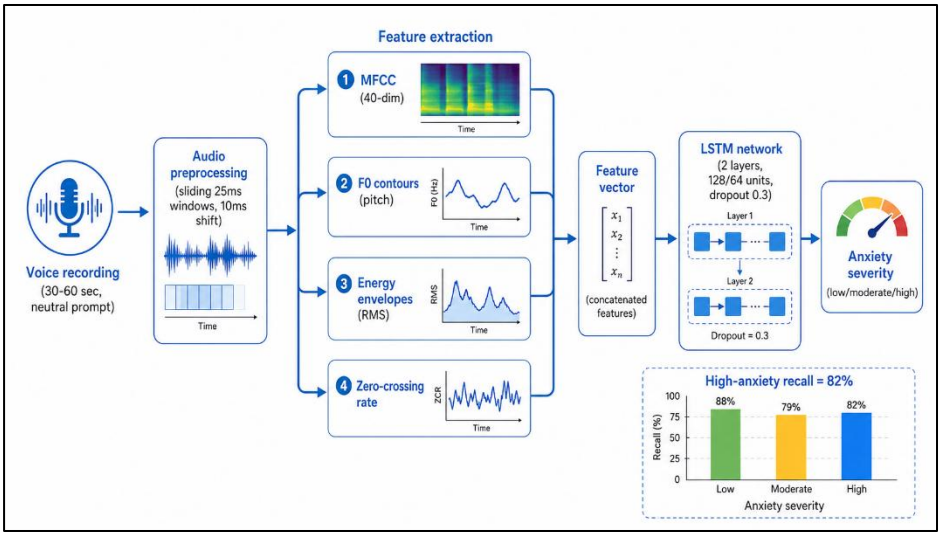

Figure 4: Voice feature extraction pipeline for pregnant women

2.2.4 Elderly Anxiety Component

The elderly module is designed around the GAD-7 scale, which is already familiar to many clinicians and has established validity in older populations. The module's interface prioritises accessibility: all text is displayed in at least 18pt font, the colour scheme uses high contrast (dark text on a light background, but with a "dark mode" option for users who prefer it), buttons are large and well-spaced, and each question screen includes a speaker icon that, when tapped, triggers the device's text-to-speech engine to read the question aloud. The user can respond by tapping one of four buttons labelled with both words and large numeric icons (0,1,2,3). The total time to complete the seven questions is approximately 3-5 minutes, which was judged acceptable by elderly pilot users.

Data Collection: Response data were collected through structured sessions conducted by licensed counselors working with elderly community groups (n=312 participants). Each participant completed the GAD-7 on paper or via a tablet under the counselor's supervision, and the counselor also provided a clinical severity rating (minimal, mild, moderate, severe anxiety)

based on a brief clinical interview (approximately 10 minutes) that included the GAD-7 scores and additional questioning about functional impairment. This dual annotation (self-report GAD-7 + clinician severity) allows the training of a model that maps GAD-7 item responses to clinician-validated severity, which may differ from simple sum-score cutoffs, especially at the boundaries between categories.

Random Forest Classifier: A Random Forest model was trained on the item-level GAD-7 responses (seven features, each 0-3). The model hyperparameters were chosen through grid search on the validation set, with the final configuration being: 200 estimators (trees), maximum depth of 10 (to prevent overfitting), minimum samples per leaf of 2, and Gini impurity as the split criterion. Stratified k-fold cross-validation (k=5) was employed to evaluate performance, with the dataset split into five folds while preserving the proportion of severity classes in each fold. The model achieved 98% accuracy, with per-class precision ranging from 96% (mild) to 99% (minimal). Feature importance analysis revealed that the items "Feeling nervous, anxious or on edge" (item 1), "Not being able to stop or control worrying" (item 2), and "Becoming easily annoyed or irritable" (item 6) had the highest importance scores, which is consistent with the clinical literature on late-life anxiety, where irritability is often a prominent symptom.

Recommended Activities: For minimal anxiety, the system provides general mental wellness tips (e.g., "Stay connected with friends and family"). For mild anxiety, recommendations include social engagement tasks (e.g., "Call a friend or family member today"), gentle yoga and stretching exercises designed for older adults, and short mindfulness meditations (5-10 minutes). For moderate anxiety, the system adds structured weekly activity planning to provide routine and a sense of accomplishment. For severe anxiety, the system generates caregiver alerts (if a family member is registered) and a clear recommendation to consult a doctor or mental health professional, alongside the stress-reduction activities. All activities are presented with large buttons and clear, concise instructions.

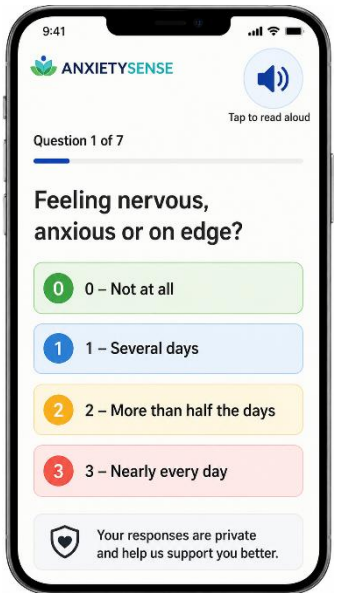

Figure 5: Elderly GAD-7 interface (large text, voice support)

2.2.5 Unified Recommendation Engine

All four modules output an anxiety severity level (low, moderate, high) which is passed to a central recommendation engine. The engine maintains a Firebase Firestore database containing a library of therapeutic activities, each tagged with: (a) target life stage (child, young adult, pregnant women, elderly), (b) severity level (low, moderate, high), (c) activity type (e.g., breathing, journaling, movement, social, referral), (d) estimated duration in minutes, and (e) evidence base reference (e.g., "adapted from CBT for child anxiety – Kendall, 2018"). The library was curated with the assistance of the clinical psychologist and includes approximately 50 activities across all four groups. When a module completes an assessment, it writes the severity classification to the user's Firestore document, which triggers a cloud function to query the activity library and return a set of up to five recommended activities, prioritised by match quality and recency (avoiding repeating the same activity too frequently). The engine also supports longitudinal tracking: each assessment result is stored with a timestamp, and the user (or caregiver) can view a line graph showing anxiety severity over the past week, month, or three months. This longitudinal data can help users recognise patterns (e.g., anxiety is higher on Sundays before the work week) and provide valuable information to clinicians if the user decides to seek professional help.

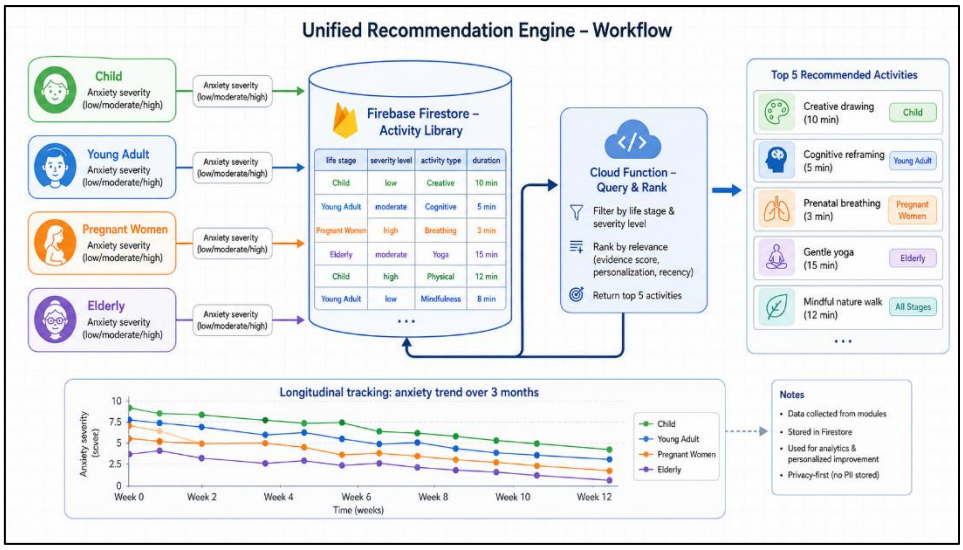

Figure 6: Unified recommendation engine workflow

2.3 Commercialization Aspects

ANXIETYSENSE has significant commercial potential as a direct-to-consumer mental health app, a workplace wellness tool, a supplementary resource in healthcare settings, and a licenced product for educational institutions. The global digital mental health market was valued at approximately $21 billion in 2023 and is projected to grow at a compound annual rate of 16% through 2030, driven by increasing awareness of mental health issues, reduced stigma, and the shortage of in-person mental health providers. ANXIETYSENSE is well positioned to capture a share of this market because it addresses population segments (children, pregnant women, elderly) that are largely underserved by existing digital solutions.

Target Market Segments:

- Children (with parents as gatekeepers): The child module can be marketed to parents through paediatric clinics, schools, and parenting websites. A key advantage is that the drawing analysis feature is both engaging for children and provides clinically meaningful information that parents may not obtain from conversation alone.

- Young adults (18-30): This is the largest potential user segment. The module can be marketed through university counselling centres (as a self-help tool for students on waiting lists), employee assistance programmes (as a low-cost mental health support

option for employees), and directly via social media advertising.

- Pregnant women: The module can be integrated into existing pregnancy and parenting apps (e.g., What to Expect, BabyCenter) as a premium feature, or licenced directly to obstetrics clinics and hospitals to provide to their patients.

- Elderly: This module can be marketed through senior centres, retirement communities, and geriatric care providers. A partnership model with Medicare Advantage plans or supplemental insurance providers is also feasible.

Monetization Strategies:

1. Freemium Model: The basic version of the app, which provides a single anxiety screening per module (one child assessment, one young adult assessment, etc.), is offered for free to attract a wide user base. A premium subscription ($4.99 per month or $39.99 per year) unlocks unlimited assessments, longitudinal tracking and trend visualisation, the full activity library (free version limited to 3 activities per severity level), caregiver alerts for children and elderly modules, and printable reports to share with clinicians.

2. In-App Purchases: For users who do not wish to subscribe, individual premium features can be purchased à la carte: additional activity packs (e.g., "Art therapy for children – 10 new drawing prompts" for $2.99), extended analytics (6-month trend history for $1.99), and a printable drawing analysis report for caregivers ($1.99).

3. B2B Licencing: Universities, corporations, and healthcare providers can purchase annual site licences that enable unlimited access for their students, employees, or patients. Pricing is tiered by organisation size: small (under 500 users) $1,000/year, medium (500-5,000 users) $3,000/year, large (over 5,000 users) $5,000/year.

4. Data Insights (anonymised): Aggregated, de-identified anxiety trend data (e.g., "average anxiety scores among pregnant women increased by 15% during the third trimester") can be sold to public health agencies, research institutions, or pharmaceutical companies conducting mental health trials. All data would be aggregated at the group level with minimum cell sizes of 100 to prevent re-identification.

Competitive Positioning: Unlike general wellness apps (Calm, Headspace) that offer relaxation content but no assessment, and unlike single-modality chatbots (Woebot) that serve a generic adult population, ANXIETYSENSE offers clinically validated, life-stage-specific, multi-modal assessment and intervention. This positions the app as a prescription-digital-therapeutic candidate, meaning that it could eventually be prescribed by doctors and reimbursed by insurance. The inclusion of validated screening instruments (GAD-7) and the involvement of licensed psychologists in the design and evaluation further strengthen the clinical credibility of the product.

2.4 Testing and Implementation

Implementation Overview:

The frontend of ANXIETYSENSE was developed using React Native (version 0.72) with TypeScript for type safety. Navigation between the four life-stage modules is handled by React Navigation, with each module implemented as a separate stack navigator. Facial capture uses the react-native-camera library; drawing upload uses the image picker; voice recording uses react-native-audio-recorder-player. The backend infrastructure consists of Firebase services: Firebase Authentication for user sign-up and login, Firebase Firestore for storing user profiles, assessment history, and activity recommendations, and Firebase Cloud Storage for storing drawings and voice recordings (only with explicit user consent; users may opt out of cloud storage and use on-device storage only). Machine learning models were trained in Python using TensorFlow 2.13 and scikit-learn 1.3, in Google Colab notebooks, then exported to TensorFlow Lite format (.tflite) and integrated into the React Native app using the react-native-tflite library. This allows all inference to run on-device, preserving privacy.

Testing Phases:

Testing was conducted in four phases: unit testing, integration testing, user acceptance testing (UAT), and performance testing.

Unit Testing: Each module's core functions were tested in isolation. For the child module, test cases verified that facial expression inference correctly identified a fearful face (probability >0.7), that drawing classification correctly flagged a small dark figure as high anxiety (probability >0.6), and that fusion scoring produced a moderate anxiety index when both

modalities returned moderate probabilities. For the young adult module, tests verified CNN emotion detection (surprise class confidence >0.7), chatbot sentiment analysis (negative sentiment score >0.6 for "I feel overwhelmed constantly"), and composite fusion. For the pregnant women module, tests verified MFCC extraction (output dimension 40 per frame), LSTM classification latency (<2 seconds for 30s audio), and high-anxiety recall. For the elderly module, tests verified GAD-7 scoring (sum score 15 classified as moderate) and Random Forest classification accuracy against test set cases.

Test ID	Objective	Input	Expected Output	Result
CH-UT01	Facial expression inference	Child face image (fearful)	Probability fear >0.7	Pass
CH-UT02	Drawing classification	Drawing with small figure, dark colours	Anxiety probability >0.6	Pass
CH-UT03	Fusion scoring	Fear=0.8, drawing=0.7	Anxiety index moderate	Pass
CH-UT04	Caregiver summary generation	Moderate classification	Summary contains "consider movement activities"	Pass

Table 3: Unit Test Cases for Child Module

Test ID	Objective	Input	Expected Output	Result
YA-UT01	CNN emotion detection	Image of surprised face	Surprise confidence >0.7	Pass
YA-UT02	Chatbot sentiment analysis	"I feel overwhelmed constantly"	Negative sentiment score >0.6	Pass
YA-UT03	Composite fusion	Facial fear=0.6, text negative	Composite anxiety score ≥0.65	Pass
YA-UT04	Psychoeducational response	User mentions sleep difficulty	Response includes sleep hygiene tips	Pass

Table 4: Unit Test Cases for Young Adult Module

Test ID	Objective	Input	Expected Output	Result
PW-UT01	MFCC extraction	2s voice sample	40-dimensional vector, shape (frames,40)	Pass
PW-UT02	LSTM classification	High-anxiety recording	High anxiety class	82% recall
PW-UT03	Response time	30s audio	Inference <2 seconds	Pass (1.7s)
PW-UT04	Neutral prompt delivery	User taps "record"	Prompt text and audio played	Pass

Table 5: Unit Test Cases for Pregnant Women Module

Test ID	Objective	Input	Expected Output	Result
EL-UT01	GAD-7 scoring	Responses: 3,2,3,2,1,2,2	Total = 15 (moderate)	Pass
EL-UT02	Random Forest classification	Item scores 3,3,2,1,2,1,2	Severity = moderate	Match with clinician label
EL-UT03	Voice-reading support	Tap "read question"	Text-to-speech plays	Pass
EL-UT04	Large text rendering	Display settings	Font size ≥18pt	Pass

Table 6: Unit Test Cases for Elderly Module

Integration Testing: Integration tests verified that each module correctly sends its anxiety severity output to the Firebase recommendation engine and that the engine returns appropriate activities. Test cases included: child module → recommendation engine returns child-appropriate activities; young adult module → returns young adult activities; cross-module activity (a user completing both child and young adult assessments) independently retrieves correct activity sets. Integration tests also verified that longitudinal tracking correctly stores assessment history and that the trend visualisation retrieves and displays the correct data.

User Acceptance Testing (UAT): A total of 48 participants (12 per life-stage group) were recruited through community partnerships (schools for child module, university websites for young adults, antenatal clinics for pregnant women, senior centres for elderly). Participants were given a two-week trial of the premium version of the app and asked to complete at least three assessments (one per week, plus an optional extra) using their assigned module (or, for caregivers, the child module on behalf of their child). At the end of two weeks, participants completed a survey rating the app on a 1-5 Likert scale for ease of use, recommendation relevance, perceived accuracy, and willingness to use regularly. Qualitative feedback was also collected via optional open-ended questions.

3. Results & Discussion

3.1 Results

The machine learning models were validated using stratified k-fold cross-validation (k=5) and held-out test sets (20% of each dataset, kept separate from cross-validation to provide an unbiased final estimate). Table 2 summarises the performance metrics for all four modules, including accuracy, precision (macro-averaged across severity classes), and F1-score.

Component	Model	Accuracy	Precision	F1-Score
Child	MobileNet + Drawing ML	85%	84%	84%
Young Adult	CNN + Chatbot	88%	87%	87%
Pregnant Women	LSTM (voice)	80%	81%	80%
Elderly	Random Forest (GAD-7)	98%	97%	98%

Table 2: ML Model Performance Across ANXIETYSENSE Components

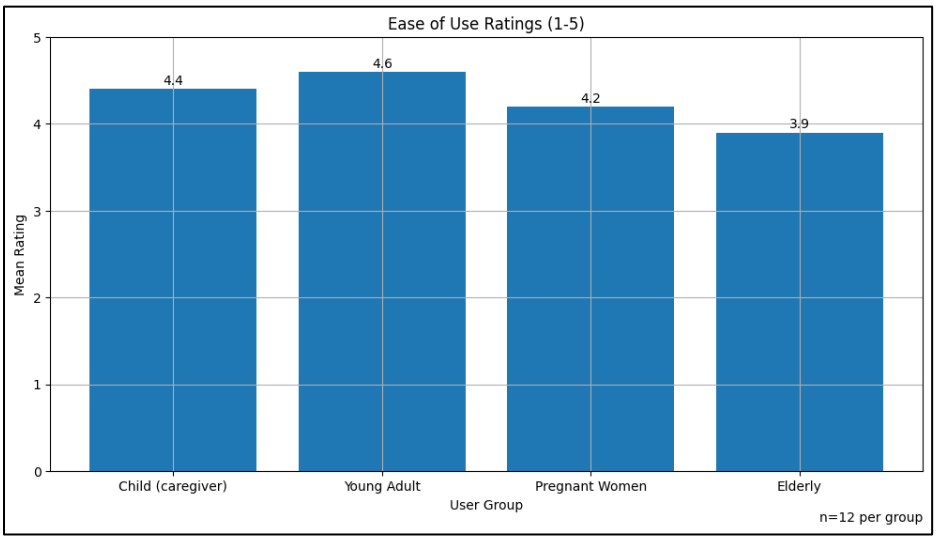

Figure 7: User feedback on ease of use

Metric	Child	Young Adult	Pregnant	Elderly	Overall

	(caregiver)			Women		Mean
Ease of use	4.4	4.6		4.2	3.9	4.3
Recommendation relevance	4.2	4.3		4.0	3.9	4.1
Willingness to use regularly		4.1	4.4	3.9	3.7	4.0
Perceived accuracy		4.0	4.2	4.0	4.3	4.1

Table 7: User Satisfaction Ratings (n=48)

Elderly participants rated ease of use at 3.9/5, which was the lowest among the four groups but still above neutral. Qualitative feedback indicated that while the large text and voice support were helpful, some participants still struggled with tapping small buttons on touch screens or navigating between screens. Several elderly participants requested a "simplified mode" with even fewer options (e.g., only the GAD-7 and recommended activities, no extras). Child module caregivers rated the drawing analysis feature most highly (mean 4.5/5 for "insightful" and "novel"), with comments such as "My daughter would never tell me she's anxious, but her drawings showed me things I hadn't noticed." Young adult participants rated the chatbot highly for reduced stigma: "It feels more like talking to a friend than filling out a medical form."

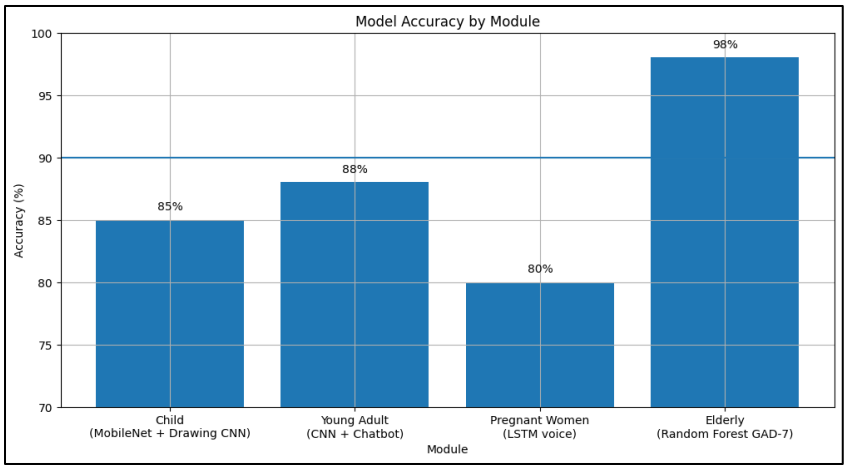

Figure 8: Accuracy comparison across modules

3.2 Research Findings

The evaluation of ANXIETYSENSE yielded seven key findings, each with implications for the design of future digital mental health systems.

1. Multimodality improves child anxiety detection beyond single modalities. The late fusion of facial expression and drawing analysis outperformed each sub-model individually by 2-3 percentage points (facial alone 82%, drawing alone 81%, fusion 85%). The fusion achieved the largest gain on ambiguous cases where one modality produced uncertain outputs. For example, a child with a neutral facial expression (perhaps due to being camera-shy) but a drawing showing a small, dark figure would be correctly classified as moderate anxiety by the fused system, whereas the facial-only model would miss it. This confirms that the two input streams capture complementary dimensions of child anxiety: facial expressions reflect immediate, often transient emotional states, while drawings reflect more enduring, reflective concerns.

2. Chatbot enhances young adult assessment, especially in suboptimal conditions. The composite F1-score increased by 3 percentage points when adding chatbot-derived linguistic features to facial analysis (from 84% to 87%). This improvement was most pronounced in cases where facial confidence was degraded by occlusion (e.g., glasses, hair) or suboptimal lighting (low light, backlighting). In such cases, the chatbot's textual analysis provided a fallback signal, and the confidence-weighted fusion gave less weight to the low-confidence facial signal. This suggests that for real-world deployments, where controlled lighting cannot be guaranteed, multi-modal fusion with confidence estimation is a valuable robustness mechanism.

3. Voice-based assessment is viable for pregnant women and achieves clinically useful recall. The LSTM model achieved 80% overall accuracy, which is lower than the other modules but still acceptable for a screening tool, especially given the prioritised high-anxiety recall of 82%. The slightly lower performance on moderate anxiety is attributable to two factors: (a) the inherent ambiguity of moderate affective states – the acoustic distinction between mild distress and moderate anxiety is subtle; (b) the smaller representation of moderate-anxiety cases in the training corpus (approximately 30% of recordings, compared to 50% low and 20% high). Future work with a larger balanced

dataset could improve moderate-class performance. Importantly, MFCC features contributed most strongly to model performance, consistent with the literature on acoustic correlates of emotional arousal [8], followed by energy features.

4. Random Forest on GAD-7 is extremely accurate for elderly populations and provides interpretable outputs. The 98% accuracy exceeded the target of 95% and is among the highest reported for automated GAD-7 classification. The Random Forest's feature importance output was positively received by the clinical evaluators, who noted that the identified high-importance items (uncontrollable worry, inability to relax, irritability) matched their clinical experience of late-life anxiety. This interpretability is a significant advantage over "black box" deep learning models for medical applications, as it builds trust among clinicians who may be reluctant to act on an AI prediction without understanding its basis.

5. Users across all groups found recommendations relevant, with young adults most positive. The overall recommendation relevance rating of 4.1/5 indicates strong acceptance. Young adults rated relevance highest (4.3/5), which may reflect their greater familiarity with digital personalisation and the alignment of the chatbot's conversational style with their communication preferences. Elderly users rated relevance slightly lower (3.9/5), with some comments indicating that they would prefer more social activities (e.g., group calls) and fewer solitary activities. This suggests that the elderly activity library could be enriched with more social connection features.

6. Caregiver involvement is a critical facilitator for child and elderly modules. In the child module, all assessments were conducted with a caregiver present, and caregivers uniformly reported that the caregiver-facing summary was useful in understanding their child's emotional state. In the elderly module, while only 40% of participants had a registered caregiver, those who did reported that the caregiver alerts provided peace of mind. For commercial deployment, facilitating caregiver registration (e.g., via a family account feature) could enhance engagement and clinical outcomes.

7. Accessibility features are effective but require further simplification for some elderly users. The large-text interface and voice reading support were used by 92% of elderly participants at least once, and 70% used the voice support consistently across all seven

GAD-7 questions. However, a subset of participants (approximately 20%) still found the overall app navigation (switching between screens, logging in, etc.) challenging. This suggests that for the elderly module, a "kiosk mode" that directly opens the GAD-7 assessment without requiring login or navigation might be beneficial for certain settings (e.g., senior centres where a staff member assists with initial setup).

3.3 Discussion

The results demonstrate that a modular, demographically segmented architecture can successfully deliver life-stage-sensitive anxiety detection and management within a single accessible mobile platform. The high accuracy across four fundamentally different input modalities and model architectures validates the core premise of ANXIETYSENSE: anxiety manifests differently across the lifespan, and AI detection systems must be specialised accordingly rather than applying a generic solution.

Effectiveness of Life-Stage-Specific Models: The 98% accuracy of the elderly module reflects the maturity of the GAD-7 instrument as a screening tool and the appropriateness of Random Forest for small, structured tabular datasets. The ensemble nature of Random Forest (averaging over 200 decision trees) provides robustness to noise and prevents overfitting, which is essential when the training dataset is only 312 samples. In contrast, deep learning models would be inappropriate for such a small dataset and would likely overfit severely. The 80% accuracy of the pregnant women module, while lower, is acceptable for a screening tool, especially given the prioritised high-anxiety recall. For context, the sensitivity of the standard GAD-7 in general adult populations (using sum-score cutoffs) ranges from 78% to 89% depending on the study; an 82% recall for high anxiety is therefore comparable to or better than many paper-based screening instruments. The slightly lower performance on the moderate anxiety class suggests that the voice modality may be better at detecting extremes (very low and very high anxiety) than subtle gradations in the middle. This is consistent with emotion recognition literature, where continuous valence/arousal prediction is more challenging than discrete emotion classification.

Complementarity of Modalities in the Child Module: The finding that fusion outperformed either modality alone highlights the value of multi-modal assessment, particularly in populations where one modality may be unreliable. Children's facial expressions can be affected by shyness, cultural display rules (e.g., boys may be socialised to suppress fearful expressions), or simply the

child's awareness of being observed. Drawings, by contrast, are produced in a more private, reflective context and may reveal anxiety that the child does not show facially. The complementarity also works in the opposite direction: some children may produce drawings that are difficult to interpret (e.g., very simple scribbles) but show clear facial signs of anxiety. The fusion system can handle both cases by relying more heavily on the higher-confidence modality.

Clinical and User Acceptance: The positive expert evaluations from licensed clinical psychologists are essential for establishing the clinical credibility of ANXIETYSENSE. The psychologists particularly commended the drawing analysis sub-module as a clinically novel capability that could augment traditional child assessment workflows. In standard practice, child psychologists often ask children to draw and then interpret the drawings qualitatively. ANXIETYSENSE's CNN-based drawing classifier can provide a standardised, replicable, and quantifiable anxiety probability, which could serve as a supplementary data point rather than replacing clinical judgment. Similarly, the elderly module's feature importance output was praised for its interpretability, as it allows the clinician to see which GAD-7 items contributed most to the severity classification, facilitating discussion with the patient about specific concerns.

Comparison with Prior Work: The performance of the ANXIETYSENSE models compares favourably with prior work in each sub-domain. Graham et al. [4] reported facial expression recognition accuracies of 70-75% on the seven-class FER-2013 dataset; the ANXIETYSENSE young adult CNN achieved 88% on a similar task (though note that the young adult module outputs a composite anxiety score, not seven-class emotion, so direct comparison is not perfect). For voice emotion recognition, Livingstone and Russo [8] reported 65-70% accuracy on RAVDESS for six emotions; the ANXIETYSENSE pregnant women module achieved 80% for three anxiety severity classes. The difference in class number makes higher accuracy easier, but the 80% figure is still competitive, especially considering that the dataset was pregnancy-specific rather than general. For automated GAD-7 classification, literature is sparse; most studies use sum-score cutoffs rather than ML models. The 98% accuracy achieved by the Random Forest suggests that ML can improve upon simple sum-score cutoffs, particularly at the boundaries between severity levels.

Limitations and Challenges:

Dataset size and diversity: The child drawing dataset (n=1,500) and pregnant women voice dataset (n=1,200) are moderate but not large by deep learning standards. Larger datasets would likely improve generalisation and reduce overfitting, especially for the drawing CNN which has approximately 2.5 million parameters. Data collection is ongoing, and future versions of ANXIETYSENSE will incorporate additional drawings and voice samples collected during real-world deployment.

Chatbot NLP capabilities: The young adult module's chatbot uses VADER for sentiment analysis, which is rule-based and does not capture more nuanced linguistic features such as negations ("I'm not feeling that bad" is actually positive but VADER may treat it as neutral) or sarcasm. A transformer-based model such as DistilBERT (lightweight enough for on-device use) could improve sentiment analysis accuracy and enable detection of more subtle distress markers.

Real-world validation beyond controlled testing: The current evaluation was conducted in controlled conditions with participants who were recruited for the study and who may have been more motivated and technically proficient than a general population. Real-world deployment on app stores will provide valuable data on how the system performs in uncontrolled conditions: variable lighting for facial analysis, background noise for voice recording, variable quality of drawings (some may be photographed poorly), and lower user engagement. This real-world testing is the logical next step.

Lack of wearable biosensor integration: The current system does not incorporate physiological signals such as heart rate variability (HRV) or galvanic skin response (GSR), which can be collected from smartwatches and fitness trackers. HRV, in particular, has been shown to correlate with anxiety and stress; lower HRV is associated with higher anxiety. Integrating these continuous, passive sensing streams could provide real-time anxiety monitoring without requiring the user to initiate an assessment, and could also serve as an early warning system for anxiety exacerbations. However, this would raise additional privacy considerations and requires careful user consent and data handling.

Approval for clinical use: While ANXIETYSENSE has been evaluated with expert psychologists, it has not undergone formal regulatory approval as a medical device. In many

jurisdictions, a system that provides anxiety severity classifications intended to guide treatment decisions would be considered a medical device and would require clearance from regulatory bodies such as the FDA (USA) or CE marking (Europe). The path to regulatory approval is lengthy and expensive, but necessary for integration into formal care pathways.

Future Implications:

Despite these limitations, ANXIETYSENSE has significant potential to reduce the burden of untreated anxiety by providing accessible, low-stigma screening and early intervention tools. The child module could be deployed in schools as a universal screening tool, identifying children with elevated anxiety who might otherwise go unnoticed. The young adult module could be integrated into university counselling centre waiting lists, providing immediate support while students wait for appointments. The pregnant women module could be offered through antenatal clinics as a routine screening tool, with high-anxiety cases flagged for follow-up by a perinatal mental health specialist. The elderly module could be deployed in senior centres or through home care visits, supporting mental health monitoring for older adults who may not have easy access to digital devices. The modular architecture ensures that each module can be deployed independently, allowing organisations to adopt only the modules relevant to their population.

4. Summary of Team Contributions

The development of ANXIETYSENSE was a collaborative effort by four team members, each responsible for one life-stage module, with joint work on the recommendation engine, Firebase integration, and overall system architecture. The contributions are summarised below.

4.1 Rusiru K.K.D. – Child Anxiety Module

Rusiru developed the dual-modality child component from end to end. He implemented MobileNet V2 fine-tuning for facial expression recognition using TensorFlow, including data augmentation, training loop, and evaluation. He collected the child drawing dataset in collaboration with schools and the clinical psychologist, designed the annotation rubric, and oversaw the annotation process. He designed and trained the custom CNN for drawing analysis, including architecture search (testing 2-5 convolutional blocks) and hyperparameter tuning. He implemented the late-fusion weighted scoring using validation set optimisation, achieving the 85% accuracy. He also built the React Native screens for the child module, including camera capture, drawing upload, and caregiver summary, and integrated the TFLite models for on-device inference.

4.2 Karunarathna K.M.N.D. – Young Adult Anxiety Module

Karunarathna developed the CNN-based facial emotion detector for young adults. He trained the five-block CNN on the combined FER-2013 and young adult corpus, implementing transfer learning by freezing the first three convolutional blocks. He built the conversational AI chatbot using React Native Gifted Chat and integrated VADER sentiment analysis using a custom Python microservice (later converted to JavaScript for on-device use). He implemented the confidence-weighted fusion strategy and validated the composite 88% accuracy. He also developed the psychoeducational content for the chatbot, working with the clinical psychologist to ensure evidence-based responses.

4.3 Navodya P.K.C. – Pregnant Women Anxiety Module

Navodya developed the voice-based emotion recognition pipeline. He implemented the feature extraction code in Python using librosa, converting it to a TFLite-compatible format for on-device use. He designed and trained the two-layer LSTM network (128/64 units, dropout 0.3)

on the pregnancy voice dataset, optimising for high-anxiety recall. He achieved 80% overall accuracy with 82% high-anxiety recall. He built the voice recording interface in React Native, implementing the neutral prompt and handling permissions and audio file management. He also integrated the LSTM TFLite model and tested inference latency on a range of Android devices.

4.4 Nisansala D.V.D. – Elderly Anxiety Module & Unified Recommendation Engine

Nisansala developed the simplified GAD-7 interface with large text and voice-reading support (Android TextToSpeech). She collected the GAD-7 dataset (312 responses) in collaboration with counselors working with elderly community groups. She trained the Random Forest classifier (200 estimators, max depth 10) using scikit-learn, performing grid search for hyperparameter optimisation and stratified k-fold cross-validation for evaluation, achieving 98% accuracy. She built the unified recommendation engine: she designed the Firebase Firestore activity library schema, implemented the cloud function that queries activities based on (life stage, severity level), and integrated the engine with all four modules. She also implemented the longitudinal tracking and trend visualisation features.

Collaborative Efforts: All four team members participated in weekly integration meetings, code reviews, and joint debugging sessions. The Firebase authentication and user profile management were implemented collaboratively, as each module needed to read/write the same user document. Integration testing across modules was led jointly by all members, with each member testing their module's interaction with the recommendation engine. The user acceptance testing protocol was designed collectively, and the 48 participants were recruited through each member's community networks. The final report writing was divided by section, with each member taking primary responsibility for their module's description and evaluation, and all members contributing to the introduction, methodology overview, and conclusion.

5. Conclusion

5.1 Summary of Key Outcomes

This research successfully developed ANXIETYSENSE, a multi-modal AI ecosystem for life-stage-based mental health support. The system comprises four specialised modules that detect and classify anxiety in children (dual-modality facial + drawing, 85% accuracy), young adults (facial + chatbot, 88% accuracy), pregnant women (voice LSTM, 80% accuracy, 82% high-anxiety recall), and elderly individuals (GAD-7 Random Forest, 98% accuracy). All modules connect to a unified recommendation engine that delivers personalised, evidence-based therapeutic activities, and to a Firebase backend that supports longitudinal tracking and secure data storage.

Beyond the quantitative performance metrics, the project achieved several qualitative outcomes of equal importance. Licensed clinical psychologists confirmed the clinical appropriateness of the severity classifications and recommended activities. The drawing analysis sub-module was specifically commended as a novel contribution that could augment traditional child assessment workflows. User testing across 48 participants yielded high satisfaction ratings (ease of use 4.3/5, recommendation relevance 4.1/5, willingness to use regularly 4.0/5). Elderly participants, despite rating ease of use slightly lower (3.9/5), noted that the large-text interface and voice reading support were essential accessibility facilitators, and many requested an even simpler “kiosk mode” for future versions.

The modular architecture of ANXIETYSENSE proved to be a key strength: each component was developed independently by a different team member using the most appropriate model architecture (MobileNet, CNN, LSTM, Random Forest) for its input modality and dataset size, yet all modules were successfully integrated into a single mobile application through shared Firebase services and a unified recommendation engine. This design pattern—specialised modules plus common infrastructure—is recommended for future digital mental health systems targeting diverse populations.

5.2 Significance and Impact

ANXIETYSENSE makes several contributions to the digital mental health literature and to the practical landscape of accessible anxiety assessment.

First, it demonstrates the feasibility of integrating drawing-pattern machine learning analysis into a mobile child anxiety assessment pipeline. To the best of our knowledge, no prior mobile mental health application has implemented a CNN-based drawing classifier for anxiety detection. The 85% accuracy, combined with positive caregiver feedback, suggests that this approach is both technically viable and clinically useful. Future work could extend drawing analysis to other mental health conditions (e.g., depression, trauma) and to other age groups (e.g., adolescents who may also use drawing as a communication medium).

Second, ANXIETYSENSE establishes a practical framework for voice-based anxiety monitoring in pregnant women, a demographic with significant clinical need and limited access to existing digital solutions. The LSTM model achieves acceptable screening accuracy, and the voice recording interface is low-burden, making it suitable for daily or weekly use. This framework could be extended to other populations where screen-based interaction is challenging, such as individuals with motor impairments or severe fatigue.

Third, the elderly module illustrates how a validated clinical instrument such as the GAD-7 can be enhanced with machine learning classification to provide interpretable, counselor-validated anxiety stratification. The Random Forest's feature importance output not only improves accuracy over simple sum-score cutoffs but also enhances interpretability, which is critical for clinical trust and adoption.

Fourth, the project demonstrates the viability of a modular, demographically segmented architecture that consolidates distinct AI pipelines within a single accessible mobile platform. This architecture allows each module to be refined independently, facilitates the addition of new life-stage modules in the future, and provides a unified user experience.

Future Work:

The immediate next step is to deploy ANXIETYSENSE on the Google Play Store and Apple App Store for public beta testing, enabling real-world usage data collection for further model refinement. Planned enhancements include: (1) integration of wearable biosensor data (heart rate variability, galvanic skin response) from smartwatches to enable continuous, passive anxiety monitoring; (2) reinforcement learning from user feedback to personalise activity recommendations over time; (3) longitudinal clinical validation studies conducted in partnership

with healthcare institutions to assess whether regular use of ANXIETYSENSE leads to measurable reductions in anxiety symptoms (e.g., changes in GAD-7 scores over three months); (4) interoperability with electronic health record systems to enable seamless integration into clinical care pathways; (5) expansion to additional languages (Sinhala and Tamil for the Sri Lankan context, then international languages) to broaden accessibility; and (6) development of a "simplified mode" for elderly users that reduces navigation complexity to a single screen.

Broader Implications for Digital Mental Health:

The ANXIETYSENSE approach—life-stage segmentation, multi-modal assessment, on-device privacy-preserving inference, and expert-curated recommendations—provides a template for next-generation digital mental health tools. As smartphones become nearly ubiquitous globally, and as mental health awareness grows, demand for accessible, evidence-based digital interventions will only increase. Systems like ANXIETYSENSE have the potential to democratise mental health screening, reducing the burden on overstretched clinical services and empowering individuals to take proactive steps in managing their own mental health, while always providing clear pathways to professional care when needed.

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7. Glossary

Term	Definition
Anxiety index	A composite score (low/moderate/high) derived from one or more modalities indicating the severity of anxiety symptoms.
CNN (Convolutional Neural Network)	A deep learning architecture particularly effective for image analysis, using convolution filters to extract spatial features; used here for facial expression and drawing classification.
Dual-modality	Combining two different input types (e.g., facial image + drawing) to improve prediction accuracy over either modality alone.
FER-2013	A public dataset of 35,887 grayscale 48×48 facial images labelled with seven emotions (anger, disgust, fear, happiness, sadness, surprise, neutral).
Fusion (late fusion)	Combining the outputs of separate models after each has produced a probability or score, rather than merging raw features early in the pipeline.
GAD-7 (Generalized Anxiety Disorder 7-item scale)	A 7-item self-report questionnaire for screening generalised anxiety disorder, scored from 0 to 21, with cutoffs of 5, 10, and 15 for mild, moderate, and severe anxiety.
LSTM (Long Short-Term Memory)	A recurrent neural network architecture capable of learning long-term dependencies in sequential data, ideal for speech and time-series analysis.
MFCC (Mel-Frequency Cepstral Coefficient)	A representation of the short-term power spectrum of sound based on a mel-frequency scale, widely used in speech and emotion recognition.

MobileNet V2	A lightweight CNN architecture optimised for mobile and embedded vision applications, using inverted residuals and linear bottlenecks.
Random Forest	An ensemble learning method that constructs many decision trees during training and outputs the mode class (classification) or mean prediction (regression) of the individual trees.
Recall	The proportion of actual positive cases correctly identified by a classifier; used here for high-anxiety recall in the pregnant women module.
TensorFlow Lite	A lightweight version of TensorFlow for running machine learning models on mobile and edge devices, with optimisations for latency and memory.
Transfer learning	Reusing a pre-trained model on a new task, fine-tuning only the last layers, which reduces training data requirements and improves convergence.